Raspberry Pi Pico 2 (RP2350)
Pins 35–40 and RUN Pin Technical Guide

Overview

Pins 35 through 40 on the Raspberry Pi Pico 2 are dedicated to the analog reference, power distribution, and USB power interface. They are not general-purpose GPIO pins but are essential for reliable hardware design.

Pin 35 – ADC_VREF

Provides the reference voltage for the 12-bit ADC. By default it is tied to the filtered 3.3 V supply. An external precision voltage reference may be connected for higher measurement accuracy.
Full-scale ADC reading (4095) corresponds to the voltage applied to ADC_VREF.

Pin 36 – 3V3(OUT)

Regulated 3.3 V output from the onboard regulator. Use this pin to power sensors, EEPROMs, displays, RTCs, and other 3.3 V peripherals. Approximately 300 mA is available for external circuitry after allowing for Pico consumption.

Pin 37 – 3V3_EN

Enable input for the onboard regulator. Normally held HIGH. Pulling this pin LOW disables the regulator and powers down the entire board, making it useful for ultra-low-power battery applications.

Pin 38 – GND

Ground reference for power and signal return.

Pin 39 – VSYS

Primary power input (approximately 1.8–5.5 V). Connect battery supplies or external regulated supplies here. The onboard regulator converts VSYS to 3.3 V.

Pin 40 – VBUS

USB 5 V input. High only when USB is connected. Useful for detecting USB presence. Avoid using it as a general external power input where USB may also be connected.

RUN Pin

The RUN pin is the RP2350 reset input (available on a test pad rather than the 40-pin header). Pulling RUN LOW resets the processor while keeping the power rails active. Releasing RUN causes the RP2350 to reboot and execute from the beginning of the program.

RUN vs 3V3_EN

Feature	RUN	3V3_EN
Function	Processor reset	Power regulator enable
Power Rails	Remain ON	Turn OFF
CPU	Resets and restarts	Powers off

RAM	Reset behaviour depends on reset type	Contents lost
Typical Use	Reset button, debugger	Ultra-low-power shutdown

Recommended Design Practice

Leave ADC_VREF at its default connection unless precision analog measurements are required. Use 3V3(OUT) to power external peripherals. Bring RUN to a pushbutton and optionally to a programming header for convenient reset during development. Expose 3V3_EN on a test pad if low-power designs are anticipated. Use VSYS as the preferred external power input and VBUS only for USB power detection.